# Huawei Access Node HSR-ACC-74 Routing Protocol Flap Resolution SOP

**Node:** HSR-ACC-74 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_7848

## Description

The routing protocol flap failure domain on Huawei Access Node HSR-ACC-74 is attributed to the inherent complexities of routing protocol interactions on this specific device role. BGP route churn and OSPF adjacency loss can be triggered by ISIS LSP flooding due to the dynamic nature of link-state routing protocols. In a scenario where the routing table is heavily loaded and the device is under high traffic conditions, the likelihood of routing protocol flaps increases. This is further exacerbated by the presence of multiple routing protocols operating simultaneously, which can lead to conflicts and instability in the control plane. The Huawei Access Node's architecture is designed to handle such scenarios, but in this specific instance, the failure domain is attributed to a combination of factors including, but not limited to, configuration inconsistencies, suboptimal routing policies, and hardware or software anomalies.

## Pre-requisite Checks

- Pre-requisite: Ensure remote hands are dispatched to HSR Layout Access to facilitate on-site intervention.

- Pre-requisite: Verify the status of redundant links to ensure they are operating within normal parameters.

## Potential Root Causes

| Cause | Probability |
|---|---|
| BGP route churn due to configuration inconsistencies | High |
| OSPF adjacency loss due to suboptimal routing policies | Medium |
| ISIS LSP flooding due to hardware or software anomalies | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: Routing table for VPN instance: 1 Network Next Hop MED Preference ----------
------------ --------- --------- 10.0.0.0/8 192.168.1.1 100 200 ,
Description: This command will display the current BGP routing table, providing insight into route churn and potential configuration inconsistencies.

Command: display ospf routing-table, Expected Output: Routing table for OSPF process: 1 Network Next Hop MED Preference ----------
------------ --------- --------- 10.0.0.0/8 192.168.1.1 100 200 ,
Description: This command will display the current OSPF routing table, enabling the identification of adjacency loss and potential suboptimal
```

```
routing policies.

Command: display isis lsp flooding, Expected Output: LSP flooding
statistics: LSPs flooded: 100 LSPs not flooded: 0 , Description: This
command will provide LSP flooding statistics, helping to identify
potential hardware or software anomalies.
```

## Resolution Steps

Step 1: Step: Check and correct BGP configuration, Command: display bgp configuration, Description: Verify BGP configuration for inconsistencies and make necessary corrections to prevent route churn.

Step 2: Step: Optimize OSPF routing policies, Command: display ospf policy, Description: Review and optimize OSPF routing policies to prevent adjacency loss and ensure optimal routing decisions.

Step 3: Step: Identify and address LSP flooding, Command: display isis lsp flooding, Description: Investigate and address LSP flooding issues to prevent hardware or software anomalies from affecting the control plane.

## Post-Check Verification

```
Command: display bgp routing-table, Expected Output: Routing table for
VPN instance: 1 Network Next Hop MED Preference ----------
------------ --------- --------- 10.0.0.0/8 192.168.1.1 100 200 ,
Description: Verify that BGP routing table is stable and accurate
after resolution.

Command: display ospf routing-table, Expected Output: Routing table
for OSPF process: 1 Network Next Hop MED Preference ----------
------------ --------- --------- 10.0.0.0/8 192.168.1.1 100 200 ,
Description: Verify that OSPF routing table is stable and accurate
after resolution.

Command: display isis lsp flooding, Expected Output: LSP flooding
statistics: LSPs flooded: 0 LSPs not flooded: 100 , Description:
Verify that LSP flooding has been resolved and is no longer affecting
the control plane.
```

## Roll-back Procedures

- Step: Restore original BGP configuration, Command: undo bgp configuration, Description: Restore original BGP configuration to prevent further route churn.

- Step: Restore original OSPF routing policies, Command: undo ospf policy, Description: Restore original OSPF routing policies to prevent adjacency loss and ensure optimal routing decisions.

- Step: Restore original LSP flooding configuration, Command: undo isis lsp flooding, Description: Restore original LSP flooding configuration to prevent hardware or software anomalies from affecting the control plane.